

# Kaushik JV

## Design System v3

*Zillennial Bridge | Deep Ink x Teal Phosphor*

The dark inversion of the Heritage Cream system. Canvas flips to Deep Ink (#1C1A14) and the accent shifts to Teal Phosphor (#A5F3FC) -- the brightest point of the teal ramp, achieving WCAG AA contrast on dark surfaces (9.8:1). CRT terminal aesthetics, inverted haptic buttons, and zero grain on dark cards.

VERSION 3.0

### TECH STACK

|            |                                                 |
|------------|-------------------------------------------------|
| Framework  | Next.js 14 (App Router) + TypeScript            |
| Styling    | CSS Custom Properties (--v3-*) + Tailwind CSS   |
| Animation  | Framer Motion -- Spring physics                 |
| Terminal   | Anonymous Pro (Google Fonts) -- monospace layer |
| Tokens     | W3C Design Tokens format (tokens.json)          |
| Repository | github.com/JVK0804/DesignSystem_Retro           |

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## 01 -- Design Philosophy

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v3 is the dark inversion of the Heritage Cream system. Every design decision is driven by the same principles as v1/v2 -- but solved for a dark canvas. The challenge: on dark backgrounds, contrast mechanics invert, glass opacity must increase, and grain becomes noise.

1

### Push to the Brightest Point

On dark surfaces, use the brightest accent in the ramp as primary. Phosphor (#A5F3FC) achieves 9.8:1 contrast on Deep Ink vs Teal Volt's 5.1:1. When the canvas is dark, you can afford brightness that would be harsh on light.

2

### Inverted Glass Logic

Dark glass needs 65% opacity (vs 45% for light) to differentiate from the background. At 45%, dark glass becomes invisible against #1C1A14. Specular drops from 32% to 8% -- white highlights are harsh on dark and should be barely perceptible.

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### No Grain on Dark

Grain is a light-scattering phenomenon -- it creates depth on cream by simulating paper texture under light. On dark surfaces, it dissolves into the dark field and adds visual noise with no perceptual benefit. GlassCard.v3 has no withGrain prop.

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### CRT Terminal Layer

Anonymous Pro monospace + scanline overlay via CSS repeating-linear-gradient. Zero canvas, zero SVG, zero JS -- pure CSS at 4px intervals with 8% black opacity.

## 02 -- Colour Tokens

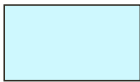
---

All tokens prefixed --v3-\*. Primary is Phosphor (#A5F3FC) -- the brightest teal, not Teal Volt. On dark backgrounds, the contrast calculation inverts: always push to the top of the ramp.

### BASE -- DEEP INK

|                                                                                     |                           |         |                          |
|-------------------------------------------------------------------------------------|---------------------------|---------|--------------------------|
|  | --v3-color-bg             | #1C1A14 | Canvas background        |
|  | --v3-color-surface        | #252118 | Raised card surface      |
|  | --v3-color-surface-raised | #2E2B22 | Further elevated surface |

## ACCENT -- TEAL PHOSPHOR

|                                                                                   |                                  |                               |                                         |
|-----------------------------------------------------------------------------------|----------------------------------|-------------------------------|-----------------------------------------|
|  | <b>--v3-color-primary</b>        | <b>#A5F3FC</b>                | Teal Phosphor -- primary accent (9.8:1) |
|  | <b>--v3-color-primary-dim</b>    | <b>#22B8CF</b>                | Teal Volt -- dimmed / hover state       |
|  | <b>--v3-color-primary-subtle</b> | <b>rgba(165,243,252,0.10)</b> | Tag background                          |
|  | <b>--v3-color-primary-border</b> | <b>rgba(165,243,252,0.24)</b> | Tag border                              |

## TEXT RAMP (ON DARK)

|                                                                                   |                                 |                |                                        |
|-----------------------------------------------------------------------------------|---------------------------------|----------------|----------------------------------------|
|  | <b>--v3-color-text</b>          | <b>#E8E4D9</b> | Heritage Cream -- primary text on dark |
|  | <b>--v3-color-text-muted</b>    | <b>#9A9488</b> | Secondary text                         |
|  | <b>--v3-color-text-disabled</b> | <b>#5A5648</b> | Disabled / placeholder                 |

## CONTRAST RATIOS (ON #1C1A14)

| Colour                    | Ratio  | WCAG | Note                         |
|---------------------------|--------|------|------------------------------|
| #A5F3FC (Phosphor) on ink | 9.8:1  | AAA  | Primary accent -- use freely |
| #22B8CF (Volt) on ink     | 5.1:1  | AA   | Dimmed state only            |
| #E8E4D9 (Cream) on ink    | 12.1:1 | AAA  | Primary text                 |
| #9A9488 (Muted) on ink    | 4.6:1  | AA   | Secondary text -- borderline |

## 03 -- Glass Tokens (Dark)

Dark glass is fundamentally different from light glass. Higher opacity (65% vs 45%), much lower specular (8% vs 32%), minimal glint (6% vs 80%). The goal is barely perceptible depth -- not the visible shimmer of the cream system.

| Token                    | Value                               |
|--------------------------|-------------------------------------|
| --v3-glass-bg            | rgba(28, 26, 20, 0.65)              |
| --v3-glass-specular-from | rgba(255, 255, 255, 0.08)           |
| --v3-glass-specular-to   | rgba(255, 255, 255, 0.01)           |
| --v3-glass-glint         | rgba(255, 255, 255, 0.06)           |
| --v3-border-glass        | 1px solid rgba(255, 255, 255, 0.10) |
| --v3-blur-glass          | blur(12px)                          |

## 04 -- CRT Terminal Tokens

The CRT terminal layer introduces Anonymous Pro monospace with scanline aesthetics. The terminal background (#0A0F0C) is darker than the canvas to create the sense of a physical screen within the interface.

| Token                  | Value                      |
|------------------------|----------------------------|
| --v3-font-terminal     | 'Anonymous Pro', monospace |
| --v3-color-crt-text    | #A5F3FC (Phosphor)         |
| --v3-color-crt-dim     | rgba(165,243,252,0.45)     |
| --v3-color-crt-bg      | #0A0F0C                    |
| --v3-size-terminal     | 13px                       |
| --v3-tracking-terminal | 0.12em                     |

#### LINE TYPE COLOURS

| Type    | Colour                  | Usage                 |
|---------|-------------------------|-----------------------|
| command | #A5F3FC (Phosphor 100%) | User input commands   |
| output  | rgba(165,243,252,0.75)  | System output         |
| error   | #FF6B6B                 | Error messages        |
| comment | rgba(165,243,252,0.45)  | Comments and metadata |

## 05 -- Glass Card Component -- GlassCard.v3

GlassCard.v3 has no withGrain prop. Grain is a light-scattering phenomenon that adds no value on dark surfaces. The component is intentionally simpler than v1/v2.

#### GLASSCARD.V3 -- USAGE

```
import { GlassCardV3 } from '@components/v3/GlassCard.v3'

// No withGrain prop -- grain is not used on dark surfaces
<GlassCardV3>
  <h3>Project Title</h3>
  <p>Description</p>
</GlassCardV3>
```

#### LAYER COMPOSITION

| # | Layer             | Description                                                       |
|---|-------------------|-------------------------------------------------------------------|
| 1 | Deep Ink canvas   | #1C1A14 base -- no grain                                          |
| 2 | Glass fill        | rgba(28,26,20,0.65) + backdrop-filter: blur(12px)                 |
| 3 | Specular ::before | linear-gradient(135deg, white/8%, white/1%) -- barely perceptible |

|   |               |                                   |
|---|---------------|-----------------------------------|
| 4 | Glint ::after | 1px top edge at 6% white opacity  |
| 5 | Content       | z-index: 20, relative positioning |

## 06 -- Haptic Button -- HapticButton.v3 (Inverted)

The primary button is inverted on dark mode. Phosphor (#A5F3FC) is a LIGHT colour -- using it as text on a dark card is low-contrast. Solution: Phosphor becomes the background and Deep Ink becomes the text. This achieves the same "glowing CTA" quality as v1/v2 but solved for the dark canvas.

| Variant   | Background         | Text          | Border  | Notes                           |
|-----------|--------------------|---------------|---------|---------------------------------|
| primary   | #A5F3FC (Phosphor) | #1C1A14 (Ink) | none    | Inverted -- light bg, dark text |
| pill      | #A5F3FC (Phosphor) | #1C1A14 (Ink) | none    | Full pill -- inverted           |
| secondary | transparent        | #A5F3FC       | #A5F3FC | Outlined -- Phosphor stroke     |
| ghost     | transparent        | #A5F3FC       | none    | Text only                       |
| disabled  | #2E2B22            | #5A5648       | none    | Raised surface + muted text     |

## 07 -- Tags & Badges -- Tag.v3

Tag.v3 uses Phosphor-based fill (teal 10%) and border (teal 24%) on dark surfaces. Text is Phosphor primary for maximum readability against the dark card surface.

| Size | Height | Font | Padding | Notes                               |
|------|--------|------|---------|-------------------------------------|
| md   | 28px   | 13px | 0 12px  | Normal case -- Phosphor text        |
| sm   | 22px   | 11px | 0 10px  | Uppercase -- +0.08em letter-spacing |

## 08 -- Terminal Block -- TerminalBlock.v3

CRT-aesthetic terminal with typewriter mode, scanline overlay, blinking cursor, and auto-scroll. The scanline effect is pure CSS -- no canvas, no SVG, zero JS. A blinking cursor uses a setInterval at 530ms in typewriter mode.

### USAGE

```
import { TerminalBlockV3 } from '@components/v3/TerminalBlock.v3'

const lines = [
  { type: 'comment', text: '# design system v3' },
  { type: 'command', text: 'npm run build' },
  { type: 'output', text: 'compiled in 240ms' },
  { type: 'command', text: 'git push origin main' },
]
```

## PROPS

| Prop       | Type           | Default | Description                                 |
|------------|----------------|---------|---------------------------------------------|
| lines      | TerminalLine[] | --      | Lines to display                            |
| prompt     | string         | ">"     | Prompt character prefix                     |
| typewriter | boolean        | false   | Stream lines one at a time (340ms interval) |
| className  | string         | "       | Additional CSS classes                      |

## TERMINALLINE INTERFACE

```
interface TerminalLine {
  type: 'command' | 'output' | 'error' | 'comment'
  text: string
}
```

## SCANLINE CSS

```
/* Zero JS, zero SVG -- pure CSS scanlines */
.v3-terminal__scanlines {
  background: repeating-linear-gradient(
    0deg,
    transparent,
    transparent 2px,
    rgba(0, 0, 0, 0.08) 2px,
    rgba(0, 0, 0, 0.08) 4px
  );
  position: absolute;
  inset: 0;
  pointer-events: none;
  z-index: 10;
}
```

## 09 -- File Structure & Usage

## DIRECTORY TREE

```
DesignSystem_Retro/
|-- styles/
|   |-- tokens.css          <- v2 base tokens
|   |-- tokens.v3.css       <- v3 tokens (--v3-*)
|-- components/
|   |-- v3/
|       |-- GlassCard.v3.tsx
|       |-- HapticButton.v3.tsx
|       |-- Tag.v3.tsx
|       |-- TerminalBlock.v3.tsx
|-- hooks/
|   |-- useHapticPress.ts   <- Shared spring hook
|-- tokens/
|   |-- tokens.json         <- W3C Design Tokens
```

## SETUP

```
// 1. Import tokens in app/layout.tsx
import '@components/styles/tokens.v3.css'

// 2. Set dark canvas in globals.css
body {
  background-color: var(--v3-color-bg); /* #1C1A14 */
  color: var(--v3-color-text);
}

// 3. Load Anonymous Pro for terminal component
import { Anonymous_Pro } from 'next/font/google'
const anonymousPro = Anonymous_Pro({ weight: '400', subsets: ['latin'] })
```

## 10 -- System Status

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| Component            | Status                       |
|----------------------|------------------------------|
| tokens.v3.css        | Complete                     |
| GlassCard.v3.tsx     | Complete                     |
| HapticButton.v3.tsx  | Complete                     |
| Tag.v3.tsx           | Complete                     |
| TerminalBlock.v3.tsx | Complete                     |
| useHapticPress.ts    | Shared -- v1/v2/v3           |
| tokens.json          | W3C format -- complete       |
| Navigation shell     | v5 (adaptive -- see v5 docs) |



